

Ryan Cho

Senior Software Engineer

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PROFESSIONAL SUMMARY

Software engineer with 7+ years designing and shipping scalable backend systems, production APIs, and data-intensive platforms across finance and capital markets technology. Proficient in REST and RPC services, SQL and NoSQL database design, and performance tuning for high-traffic workloads. Known for cross-team collaboration, reducing technical debt, and establishing patterns that improve reliability and long-term maintainability.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected and delivered **REST and gRPC services** supporting portfolio analytics workflows, enabling sub-200ms response times across 50+ internal endpoints.
- Led migration of legacy monolith modules into **containerized microservices** on Kubernetes, reducing deployment cycles from weekly to daily releases.
- Redesigned **PostgreSQL and Redis data layers** with optimized query patterns and caching strategies, cutting p95 read latency by 40% on reporting dashboards.
- Drove **technical debt reduction initiative** across 3 service domains, refactoring authentication and authorization flows to improve auditability and maintainability.
- Partnered with infrastructure teams to establish **CI/CD pipelines** and blue-green deployment standards, improving production rollout reliability and rollback speed.
- Built **event-driven ingestion pipelines** processing daily market and position data feeds, ensuring idempotent processing and data consistency at scale.
- Mentored 2 junior engineers on **backend design patterns**, code review practices, and production debugging, accelerating onboarding and team velocity.
- Implemented **distributed tracing and structured logging** across core services, reducing mean time to diagnosis for production incidents by 35%.
- Collaborated with product and risk stakeholders to define **API contracts** and service boundaries for a new investor portal, aligning delivery with compliance requirements.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Developed and maintained **mission-critical REST services** supporting trading operations and regulatory reporting for global banking platforms.
- Optimized **SQL query performance** across Oracle and PostgreSQL schemas handling millions of daily transactions, reducing batch job runtime by 30%.
- Contributed to **microservices decomposition** of legacy Java applications, improving modularity and enabling independent team ownership.
- Implemented **secure API authentication** and role-based access controls meeting bank-grade security and compliance standards.
- Supported **production incident response** and root-cause analysis for distributed systems, coordinating with operations and infrastructure teams.
- Built **data validation pipelines** for trade lifecycle events, catching anomalies before downstream settlement processing.
- Automated deployment workflows with **Docker and Jenkins pipelines**, standardizing release processes across 4 application teams.

- Participated in **architectural design reviews** for new platform capabilities, advocating for scalable patterns and operational observability.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Built **Node.js REST APIs** supporting mobile payment reconciliation workflows for restaurant clients across the New York metro area.
- Implemented **PostgreSQL schemas** and migration scripts for transaction history storage, enabling reliable audit trails for finance teams.
- Developed **React dashboard components** surfacing real-time check and payment status, improving operational visibility for internal users.
- Wrote **unit and integration tests** covering critical API endpoints, increasing coverage and reducing regression risk during feature releases.
- Collaborated with senior engineers on **code reviews and debugging** production issues, gaining hands-on experience with full-stack delivery practices.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Developed **responsive web pages** for a nonprofit fundraising platform using HTML, CSS, and JavaScript.
- Integrated **third-party donation APIs** to streamline payment collection and donor receipt workflows.
- Improved **page load performance** through asset optimization and caching, delivering faster experiences on mobile devices.
- Coordinated with stakeholders to translate **design requirements** into maintainable front-end components within a tight 3-month delivery window.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, HTML5, CSS3, JavaScript

Backend: Python, Java, Node.js, REST APIs, gRPC, Microservices

Databases: PostgreSQL, Oracle, MongoDB, Redis, SQL Query Optimization

Cloud & DevOps: Kubernetes, Docker, AWS, Jenkins, CI/CD, Terraform

Monitoring & Tools: Prometheus, Grafana, Datadog, ELK Stack, Git

Workflow & PM: Agile/Scrum, Jira, Cross-functional Collaboration, Technical Documentation